International Journal of Rock Mechanics & Mining Sciences 161 (2023) 105299



Contents lists available at ScienceDirect

International Journal of Rock Mechanics and Mining Sciences

journal homepage: www.elsevier.com/locate/ijrmms





Investigation into shale softening induced by water/CO_2-rock interaction

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ARTICLE INFO

Keywords:
Shale softening
Water/CO_2-Rock interaction
Hydration
Nanoindentation
Micromechanical parameter

ABSTRACT

To investigate the shale softening behavior induced by water/Carbon dioxide (CO_2)-rock interaction, a series of scanning electron microscope (SEM) observation and nanoindentation tests were performed on carbonate-rich shale specimens after high-temperature and high-pressure static soaking treatments. The characteristics of shale surface softening and the evolution of micromechanical properties were analyzed. It was indicated that water/CO_2-rock interaction resulted in different extents of micromechanical degradation, which was closely related to both the type of soaking fluid and the soaking time. According to the shale surface observation by SEM images, the aperture of hydration-induced fractures after soaking with water was larger than that of adsorption-induced fractures after soaking with CO_2. The statistical results of nanoindentation tests show that when soaking with water, the Young's modulus and the hardness ultimately decreased by 34.6% and 57.6% with the soaking time increased from 2 to 6 d, respectively. When soaking with CO_2, the extent of shale softening was alleviated. After soaking for 6 d, the ultimate degradation extents were 14.8% and 30.0% for the Young's modulus and the hardness, respectively. Furthermore, due to the higher penetrability of CO_2, the micromechanical degradation of the shale surface mainly occurred in the first 2 d. Indents located in the vicinity of shale laminas or induced fractures tended to possess the lower Young's modulus and the hardness than those located in shale matrix, meanwhile resulting in crushed zones or fracture compaction.

1. Introduction

With the worldwide increasing demand for fossil energy resources, shale oil has become a hot field of exploration and development of unconventional oil and gas resources.[1,2] Hydraulic fracturing is utilized to create large-scale complex fracture networks in ultra-tight unconventional formations to provide flow channel systems for the extraction of shale oil.[3–5] The long-term conductivity of the hydraulic fracture networks significantly influences the performance of shale oil wells.[6,7] Although a large amount of proppant is transported into the formation to prevent the closure of hydraulic fracture networks, the fracture conductivity inevitably decreases due to the embedment and crushing of the proppant pack under high confining pressure. This phenomenon is closely related to the mechanical properties of shale surfaces and proppant particles. Previous studies suggest that when the massive fracturing fluid is injected into and retained in the shale formation, fluid-rock interaction occurs.[8,9] Consequently, both the physical and mechanical properties of shale tend to be changed, which will further influence the long-term effectiveness of hydraulic fracture networks. Thus, it is of great significance to investigate the effect of fluid-rock interaction on the mechanical properties of shale.

Generally, the mechanisms of fluid-rock interaction vary with the types of fracturing fluids. To date, water-based fracturing fluid is still widely used for the stimulation of unconventional reservoirs. However, considering the benefits in formation protection, improvement of hydraulic fracture network complexity, oil recovery enhancement, greenhouse effect reduction, etc., carbon dioxide (CO_2) has been regarded as a promising alternative fracturing fluid and used in the development of shale oil reservoirs.[5,10,11] For the water-based fracturing fluid, the hydration of clay minerals (such as smectite and illite/smectite mixed

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https://doi.org/10.1016/j.ijrmms.2022.105299
Received 23 July 2022; Received in revised form 15 October 2022; Accepted 6 December 2022
Available online 10 December 2022
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layer) causes shale swelling.[12,13] Experimental results show that the shale swelling tends to alternate the pore structure and induced new fractures, consequently resulting in a reduction in mechanical strength.[14,15] Zhong et al.[16] pointed out that compared with the calcite-rich Wufeng shale, the mechanical properties of clay-rich Longmaxi shale showed stronger sensitivity to the water-rock interaction. For the CO_2-based fracturing fluid, besides the CO_2 adsorption-induced shale swelling, the mechanisms of CO_2-rock interaction also include the chemical dissolution of carbonate minerals, etc.[17–20] These alterations of physical and mechanical properties of shale after CO_2 exposure was dependent on not only the microstructure and mineral composition of shale but also the phase state of CO_2. When the temperature and the pressure of CO_2 exceed 31.3 °C and 7.38 MPa, respectively, it becomes the supercritical state. Pan et al.[21] suggested that supercritical CO_2-shale interaction had a more significant influence on the shale pore structure alteration as well as porosity and permeability enhancement than subcritical CO_2-shale interaction due to the greater solubility and expansibility of the supercritical state. Similar conclusions were also reported by Okamoto et al.[22] and Zhou et al.[23] Consequently, supercritical CO_2 tended to cause more severe mechanical degradation of shale than subcritical CO_2.[23,24]

In most previous studies, the influence of fluid-rock interaction on the macro mechanical properties of shale has been systematically investigated, while studies focused on the micromechanical properties of shale were insufficient.[14,18,20] In fact, the micromechanical properties of shale are more reasonable to quantify the fluid-rock interaction-induced shale surface softening.[25–27] Recently, the application of the micro/nanoindentation testing technique provides an effective way to investigate the mechanical properties of shale from the micro-scale aspect.[28–31] Song et al.[32] investigated the softening rate of Longmaxi shale caused by slick water-shale interaction under coupled thermo-hydro-mechanical-chemical conditions. Lu et al.[33] suggested that the shale softening advancement was intrinsically controlled by permeability. Liu et al.[34] and Song et al.[9] further studied the hydration-induced creep behavior of shale. Experimental results show that the shale softening caused larger creep deformation and creep rate. Additionally, Shi et al.[35] characterized the micromechanical property alteration of shale after supercritical CO_2 treatment and found that the decrease of Young's modulus and hardness was more significant in clay-rich areas than in quartz-rich areas. Li et al.[20] revealed the chemical effect of CO_2 solution on the micromechanical strength of carbonate-rich shale. Even though these abovementioned studies have been reported, the micro-scale softening behavior of the shale was mainly investigated after soaking with a certain type of fluid, i.e., water or CO_2. The differences between shale softening behavior induced by water/CO_2-rock interaction remain unclear.

This paper aims to investigate the influence of water/CO_2-rock interaction on softening behavior of shale. Static soaking treatment was performed on carbonate-rich terrestrial shale specimens from the Shengli oil field under high-temperature and high-pressure conditions. After soaking for different durations, the scanning electron microscope (SEM) analysis was performed to observe the characteristics of shale surface softening. Then, the grid nanoindentation testing technique was introduced to reveal the micromechanical degradation of the shale surface. The influence of the type of soaking fluid and the soaking time on the micromechanical properties of shale was discussed in detail based on the statistical analysis and the deconvolution method. Furthermore, three typical types of location-dependent indents were classified by combining the geometry of residual indents and the micromechanical response with the indentation depth. The experimental results can give some new insights into understanding the performance evolution of shale oil/gas wells and may provide a new perspective for optimizing the usage of aqueous/non-aqueous fracturing fluids.

2. Experimental method

2.1. Specimen preparation and static soaking treatments

The shale used in this study were down-hole cores collected from the lower part of the Shahejie Formation, Zhanhua Sag, Jiyang Depression, Shengli oil field. The true vertical depth of down-hole cores ranges from 3080.75 to 3126.95 m. According to the X-ray diffraction tests, the mineral components of the shale specimens consisted of calcite 71.4%, quartz 9.0%, dolomite 6.7%, clay mineral 6.6%, and pyrite 3.9%. Among clay minerals, the illite/smectite mixed layer accounts for approximately 60.8%, and illite 34.8%. The cubical specimens used for the nanoindentation test were prepared with the dimension of 1.5 cm × 1.5 cm × 1.0 cm. In particular, the direction of height was parallel to the bedding planes. For each specimen, the end faces of 1.5 cm × 1.5 cm were mechanically polished considering the surface smoothness and flatness requirement of nanoindentation tests.

In this study, to simulate the water/CO_2-shale interaction under high-temperature and high-pressure conditions, the soaking temperature and pressure were set at 80 °C and 20 MPa, respectively. Under this condition, CO_2 is in the supercritical state. The static soaking treatments were performed using a high-temperature and high-pressure soaking device, as shown in Fig. 1. This device was mainly composed of a soaking tank made of Hartz alloy, a high-temperature oven, two 2000 mL piston vessels, a constant rate and constant pressure syringe pump, heating pipelines, and a CO_2 source. The procedures of static soaking treatment can be summarized as follows: (1) The shale specimen was firstly fixed in the soaking tank which was filled with water/CO_2. Noting that, when soaking with CO_2, the soaking tank was filled to the initial pressure of 5 MPa (equal to that of the CO_2 source). (2) The soaking tank was heated to the target temperature of 80 °C in the high-temperature oven to simulate the underground condition. (3) The soaking tank was further pressurized to 20 MPa using the syringe pump. This target pressure was determined according to the difference between the bottomhole pressure and the pore pressure during the early stage of the shut-in treatment. (4) Ensure that the soaking tank was well sealed and maintained the high-temperature and high-pressure conditions for 2, 4, and 6 d, respectively.

2.2. Nanoindentation tests

To quantitatively investigate the shale softening behavior after different static soaking treatments, grid nanoindentation tests were performed to obtain the micromechanical parameters (including the Young's modulus and the hardness) using a Keysight G200 nanoindenter,[26,27] as shown in Fig. 2a. The rectangular areas for grid nanoindentation tests were selected with no obvious original fractures. Considering the data sufficiency, a 7 × 7 matrix indentation grid scheme was assigned for each shale specimen (Fig. 2b). To prevent the potential interference among the adjacent indents, the interval between the two adjacent indents was 100 μm (much greater than the maximum indentation depth of 4000 nm).

In this study, the continuous stiffness measurement (CSM) method was used so that the continuous evolution of micromechanical parameters with indentation depth was determined.[30,36] Fig. 2c and d shows the loading history and typical load-indentation depth curves, respectively. The testing process of the CSM method can be divided into five stages (Fig. 2d). During the loading stage (OA), the load was gradually applied at a constant strain rate of 0.05 s^{-1} and the harmonic displacement target was 2 nm with a frequency of 45 Hz. When the indentation depth reached the target value (4000 nm), the maximum load was maintained constant. The holding stage (AB) lasted for 10 s, during which the indentation further increased due to the creep of indents. After that, the unloading stage (BC) was taken at a constant unloading rate until the residual load was 10% of the maximum value. A second holding stage (CD) lasted for 75 s to record the drift of

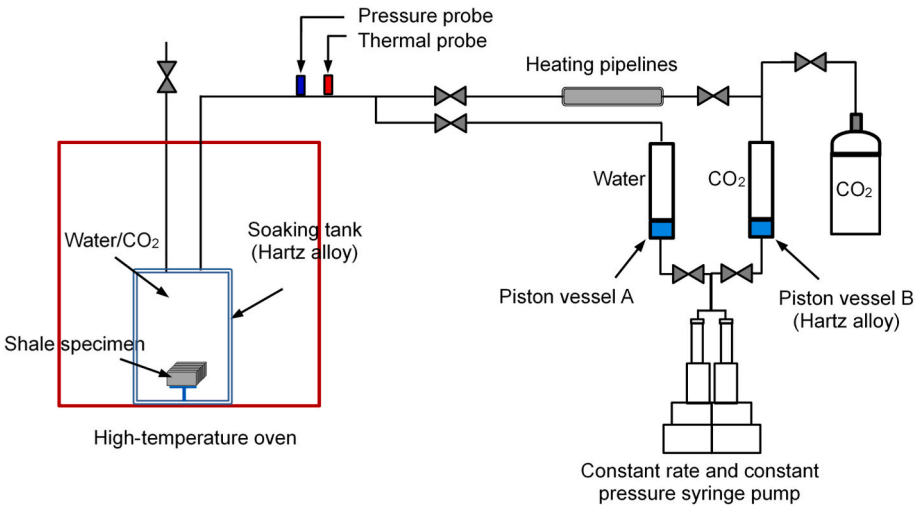


Fig. 1. The scheme of static soaking device (After Zou et al.[18]).

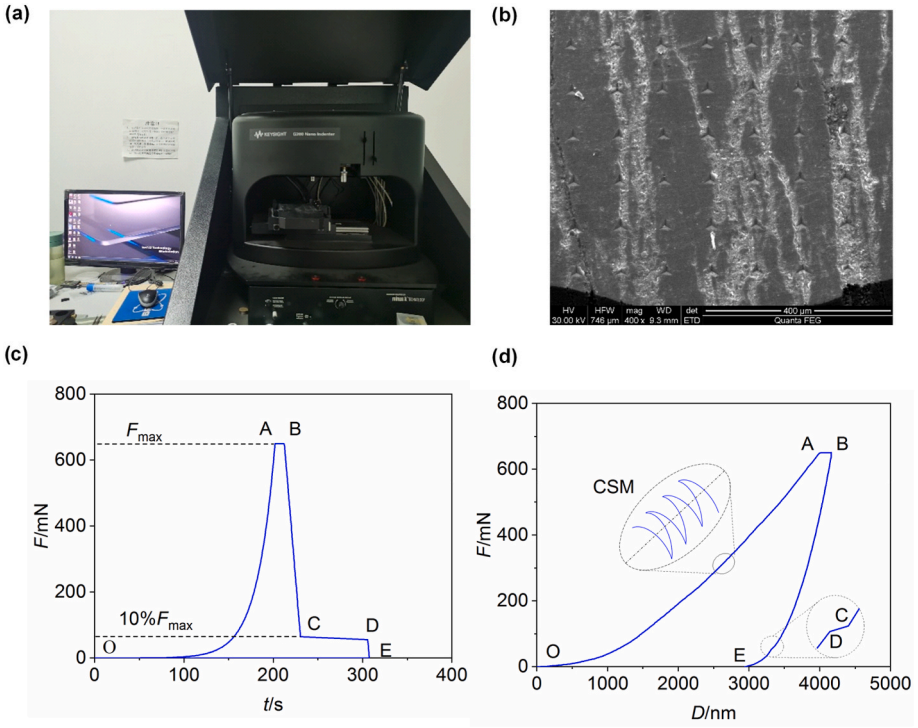


Fig. 2. Nano indentation test: **a** G200 Nanoindenter; **b** SEM image of grid indents; **c** Loading history; **d** Typical load-indentation depth curve.

temperature. Finally, the residual load was linearly removed in the unloading stage (DE).

For the CSM method, the contact stiffness (S) can be illustrated as[37]

$$S = \left[\frac{1}{(F_0/z_0)\cos\varphi - (K_s - Mw^2)} - \frac{1}{K_f}\right]^{-1} \tag{1}$$

where z_0 and w is the amplitude and frequency of the indenter oscillation, respectively. F_0 is the amplitude of the load. φ is the phase angle by which the displacement lags load. K_s, K_f and M are parameters related to the experimental setup.

During the loading stage, the hardness (H) of shale specimens can be determined by combining the following equations[38,39]

$$h_c = h_{max} - 0.75\frac{P_{max}}{S} \tag{2}$$

$$A_c = \frac{3\sqrt{3}\tan\theta}{\cos\theta} h_c{}^2 \tag{3}$$

$$H = \frac{P_{max}}{A_c} \tag{4}$$

where h_c is the contact depth, h_{max} and P_{max} are the maximum contact depth and the maximum load in each circle. A_c is the contact area. θ is the constant related to the geometry of the indenter ($\theta = 65.35°$ for the Berkovich indenter).

Additionally, the Young's modulus (E) of the shale specimens can be calculated by[38,39]

$$\frac{1}{E_r} = \frac{1-\nu^2}{E} + \frac{1-\nu_i{}^2}{E_i} \tag{5}$$

$$E_r = \frac{\sqrt{\pi}}{2\alpha}\frac{S}{\sqrt{A_c}} \tag{6}$$

where E_r is the reduced modulus ($\alpha = 1.034$ for the Berkovich indenter). E_i and ν_i are the Young's modulus and the Poisson's ratio of the intender ($E_i = 1141$ GPa and $\nu_i = 0.07$ for the Berkovich indenter), respectively. ν is the Poisson's ratio of the shale specimens.

On the basis of a large number of grid indents, the statistical deconvolution method is generally used to identify the mechanical behavior of different phases, i.e., stiff, medium stiff, and soft phases, in shale.[29,40] For a certain phase i, the normal distribution can be described as

$$f_i(x|\mu_i,\sigma_i) = 1\Big/\sqrt{2\pi}\sigma_i \exp\left[-\frac{(x-\mu_i)^2}{2{\sigma_i}^2}\right] \tag{7}$$

where $f_i(x|\mu_i,\sigma_i)$ is the probability density function of phase i. μ and σ_i are the expected value and the standard deviation of the micromechanical parameter of phase i, respectively.

Considering the proportion of each phase, the total probability density function ($f(x)$) can be written as[15,41]

$$f(x) = \sum_{i=1}^{3} c_i f_i(x|\mu_i,\sigma_i) \tag{8}$$

$$\sum_{i=1}^{3} c_i = 1 \tag{9}$$

where c_i is the proportion of phase i.

In this study, the optimal values of the unknowns for each phase's Gaussian distribution were determined by fitting the total probability density function with the experimental data using the least-squares method.[41,42]

3. Experimental results and analysis

3.1. Shale surface observation after soaking treatments

After different static soaking treatments, SEM observation was performed to reveal the surface softening characteristics of shale specimens (Fig. 3). When soaking with water, hydration-induced tensile fractures can be observed as shown in Fig. 3a–d. Meanwhile, the escape and migration of clay minerals tended to generate some new pores as the red circles indicated. It was also found that both the aperture and the number of hydration-induced fractures significantly increased with the soaking time. According to Fig. 3a and b, the aperture of typical hydration-induced fractures enlarged from 1.43 to 2.23 μm (by approximately 56%) as the soaking time increased from 2 to 4 d. After soaking for 6 d, wider hydration-induced fractures with the apertures of 2.45 and 4.73 μm were observed in the SEM images, which were also accompanied by some narrow secondary fractures and stripped debris of clay mineral (Fig. 3c and d).

By comparing the SEM images of shale surfaces after soaking with CO_2 and water, one could observe that the aperture of CO_2 adsorption-induced fractures was smaller than that of hydration-induced fractures. Meanwhile, as shown in Fig. 3e–g, the aperture of CO_2 adsorption-induced fractures at the widest points increased from 0.41 to 1.75 μm with the soaking time increasing from 2 to 6 d. The obvious variation of fracture aperture may be resulted from the falling off of local particles during depressurization (Fig. 3f and g). Additionally, when CO_2 was dissolved into water inside the laminas, carbonic acid was formed.[43,44] Consequently, carbonate minerals could be dissolved, which was beneficial for pore generation and enlargement.[20] Fig. 3h shows the dissolution pores observed around the carbonate minerals.

In summary, the experimental results of SEM observation indicated that the damage extent of the shale surface was closely related to not only the type of soaking fluid but the soaking time. The microstructure of shale specimens after soaking with water changed more significantly than those after soaking with CO_2. Furthermore, compared with the CO_2-rock interaction mechanism, the water-rock interaction mechanism tended to be more sensitive to the soaking time in terms of the shale softening behavior in this study. This can be attributed to the strong penetrability of supercritical CO_2. The ultra-low interfacial tension and viscosity of supercritical CO_2 make it easier to penetrate the shale micro pores and laminas.[5,11] Thus, the soaking time tended to influence the depth extent of CO_2-shale interaction, while having little influence on the damage extent of the shale surface due to the adsorption equilibrium in the vicinity of the fracture surface.

3.2. Micromechanical properties

The nanoindentation test is an effective technique for the quantitative characterization of micromechanical properties and has been widely used in previous studies.[35–37] Fig. 4 shows the typical results of the CSM nanoindentation tests. To investigate the influence of different soaking conditions on the variation of micromechanical parameters with the increasing indentation depth, the Young's modulus and the hardness of shale specimens at indentation depth from 1000 to 4000 nm with an interval of 500 nm were analyzed. Fig. 5 demonstrates the evolution tendency of statistical micromechanical parameters with indentation depth after different soaking treatments.

It was found in Fig. 5 that as the indenter gradually penetrated the surface of the original shale specimen, the statistical micromechanical

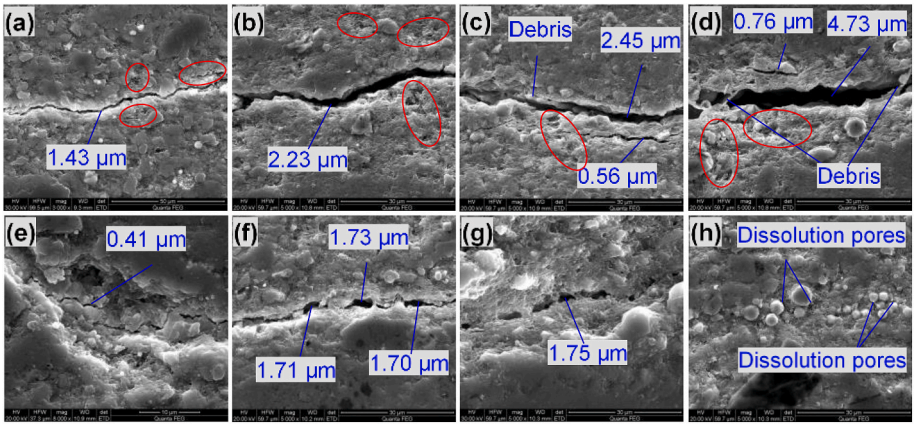


Fig. 3. SEM images of shale specimens after different soaking treatment: **a** water-2 d; **b** water-4 d; **c** and **d** water-6 d; **e** CO_2-2 d; **f** CO_2-4 d; **g** CO_2-6 d; **h** local dissolution pores.

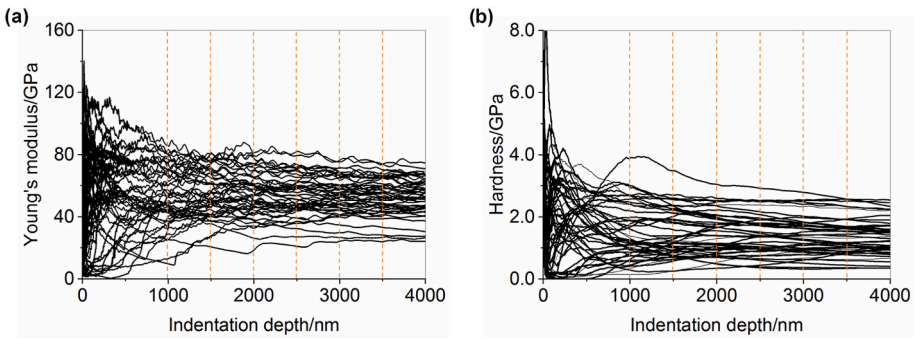


Fig. 4. Continuous variation of micromechanical parameters with indentation depth: **a** Young's modulus; **b** Hardness.

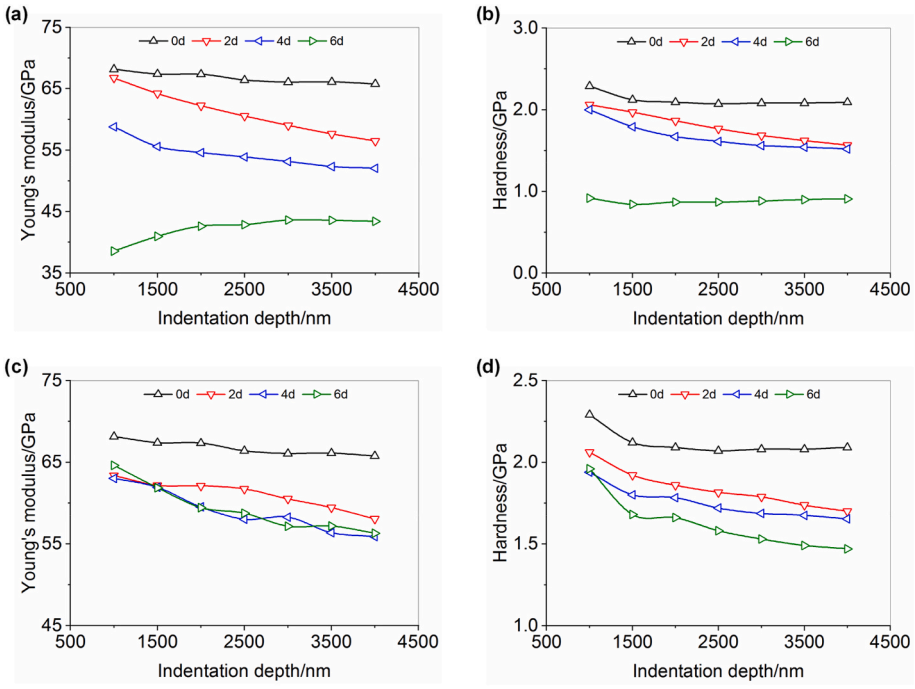


Fig. 5. Micromechanical properties evolution with indentation depth after different soaking treatments: **a** and **b** Young's modulus and hardness after soaking with water; **c** and **d** Young's modulus and hardness after soaking with CO_2.

parameters slightly decreased and then became stable. After soaking with water, both the Young's modulus and the hardness of shale specimens significantly decreased due to the hydration effect. The longer the soaking time, the more severe degradation of micromechanical parameters at the same indentation depth (Fig. 5a and b). Additionally, it is worth noting that when the soaking time was 6 d, the Young's modulus of the shale specimen showed an obvious increasing tendency before the indentation depth of 2500 nm and the hardness showed a slight increase when the indentation depth exceeded 500 nm, as the green lines shown in Fig. 5a and b. As the measurement value of hardness was generally small, the increasing tendency of hardness with the increasing indentation depth was limited. This abnormal increasing phenomenon was closely related to the significant enhancement of hydration-induced damage, which will be further discussed according to the micromechanical response of indents located in different areas in the following text.

When soaking with CO_2, the hardness of shale specimens decreased as the soaking time increased while the Young's modulus changed little after soaking for 4 d and 6 d (Fig. 5c and d). However, it was found that the degradation extent of micromechanical parameters at a series of indentation depths was smaller than that of soaking with water. This agreed well with the comparison results of microstructure damage extent by SEM observation (Fig. 3).

As the indentation depth increased, the contact area enlarged. Consequently, the micromechanical parameters at the larger indentation depth could better reflect the combined effect of mineral distribution and the damage caused by water/CO_2-shale interaction. Herein, the average values of the Young's modulus and the hardness between 3500 nm and 4000 nm were determined as the ultimate micromechanical parameters for each shale specimen. Taking the Young's modulus for instance, the frequency distribution characteristics were discussed based on the deconvolution analysis method, as shown in Fig. 6. According to Luo et al.,[36] the bin size, i.e., the statistical interval, has a great influence on the deconvolution analysis results. Herein, the proper bin size was determined as 4 GPa so that not only the noise among the random measurement was reduced, but the true frequency distribution characteristics were revealed. The nanoindentation data was generally deconvolved into three phases in this study. It should be noted that due to the existence of induced fractures after different soaking treatments, the mechanical phases in this carbonate-rich shale mainly depended on the microstructure characteristics of the location of indents. The total curves showed the total probability density distribution of the micromechanical parameter, which was obtained by considering the proportion of each phase.[45] Meanwhile, the evolution of micromechanical

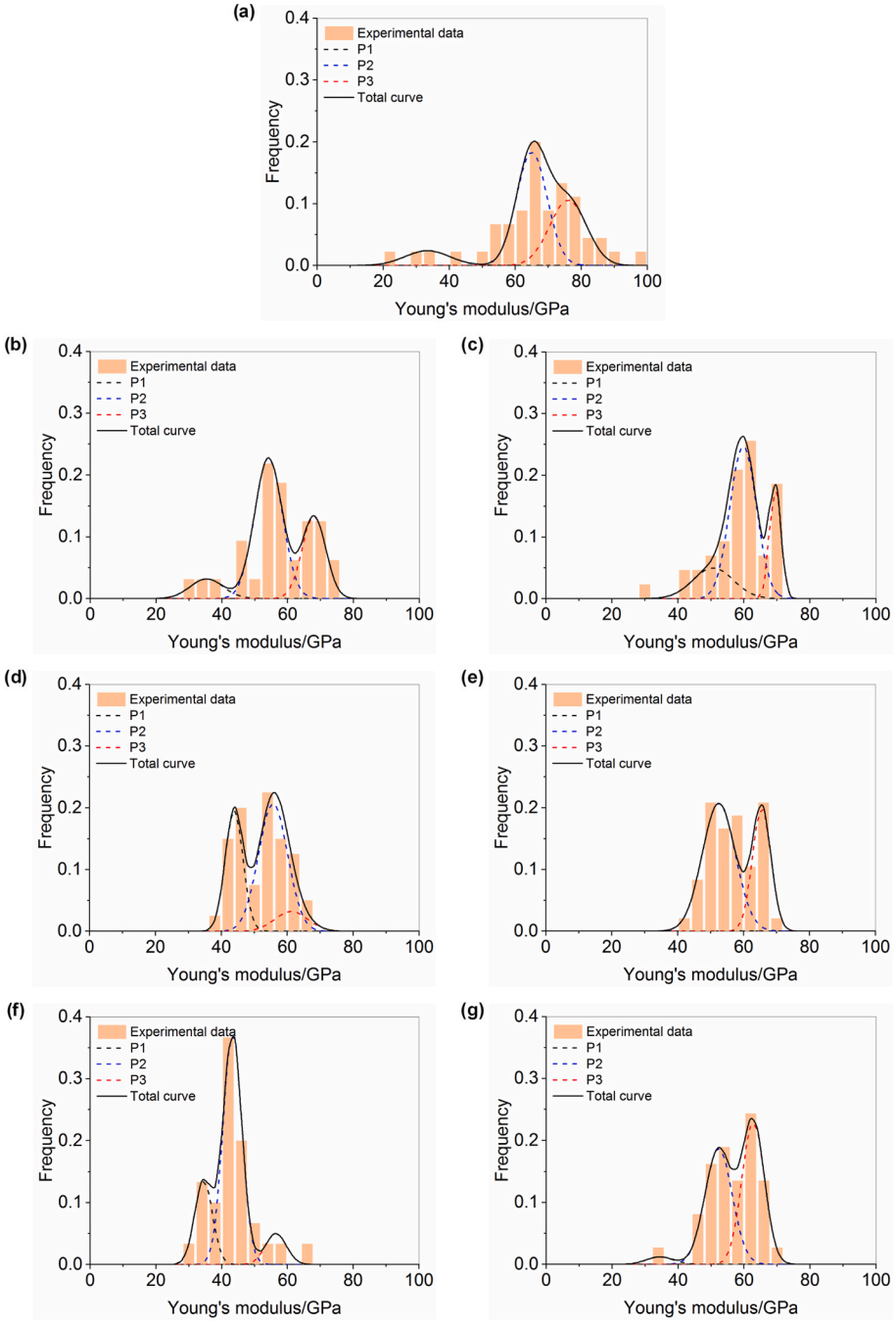


Fig. 6. Deconvolution analysis of Young's modulus distribution after different static soaking treatments: **a** Original specimen; **b** water-2 d; **c** CO_2-2 d; **d** water-4 d; **e** CO_2-4 d; **f** water-6 d; **g** CO_2-6 d.

parameters with soaking time was shown in Fig. 7.

According to the analysis results of deconvolution in Fig. 6, the statistical characteristics of Young's modulus were mainly dominated by two mechanical phases. The experimental results of nanoindentation tests showed that the Young's modulus of the original shale specimen was mainly distributed in the range of 52–80 GPa (Fig. 6a). The total curve of probability density distribution was characterized by a single peak at approximately 66 GPa due to the overlap of two mechanical phases (P1 and P2 in Fig. 6a). After different soaking treatments, two dominated peaks tended to be observed in the total curves. By comparing Fig. 6b, d, and 6f, it was found that when soaking with water, the two peaks in total curves moved leftward as the soaking time increased. Two dominant peaks of the Young's modulus were 54/68, 44/56, and 34/44 GPa after soaking for 2, 4, and 6 d, respectively. This indicated that the shale softening degree aggravated with the increasing soaking time due to the water-shale interaction. However, compared with that soaking with water, the shale softening degree after soaking with CO_2 was less severe. As shown in Fig. 6c and e, the two dominant peaks were 60/70 and 52/66 GPa after soaking for 2 and 4 d, respectively. Thereafter, it was found that the statistical distribution of Young's modulus changed little (to 52/64 GPa) when the soaking time further increased to 6 d by comparing Fig. 6e and g.

Fig. 7a shows that the Young's modulus linearly decreased with the soaking time after soaking with water. Compared with the original shale specimen (66.30 GPa), the Young's modulus decreased to 57.32, 52.00, and 43.34 GPa (i.e., by approximately 13.5%, 21.6%, and 34.6%) after

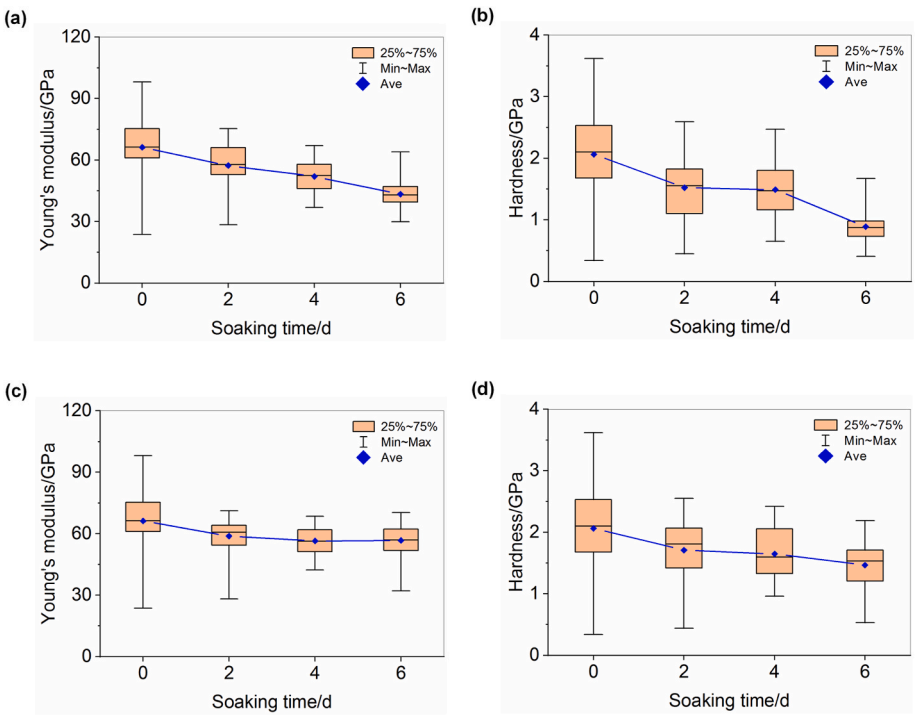


Fig. 7. Micromechanical properties evolution after soaking for different times: **a** and **b** Young's modulus and hardness after soaking with water; **c** and **d** Young's modulus and hardness after soaking with CO_2.

soaking with water for 2, 4, and 6 d, respectively. Meanwhile, the hardness decreased from 2.10 GPa to 1.52 and 1.49 GPa (a similar decline of 27.6% and 29.0%) after soaking 2 and 4 d, while to 0.89 GPa with a large decline of 57.6% when the soaking time further increased to 6 d, as shown in Fig. 7b. Differently, it was indicated that the degradation of micromechanical strength mainly occurred during the first 2

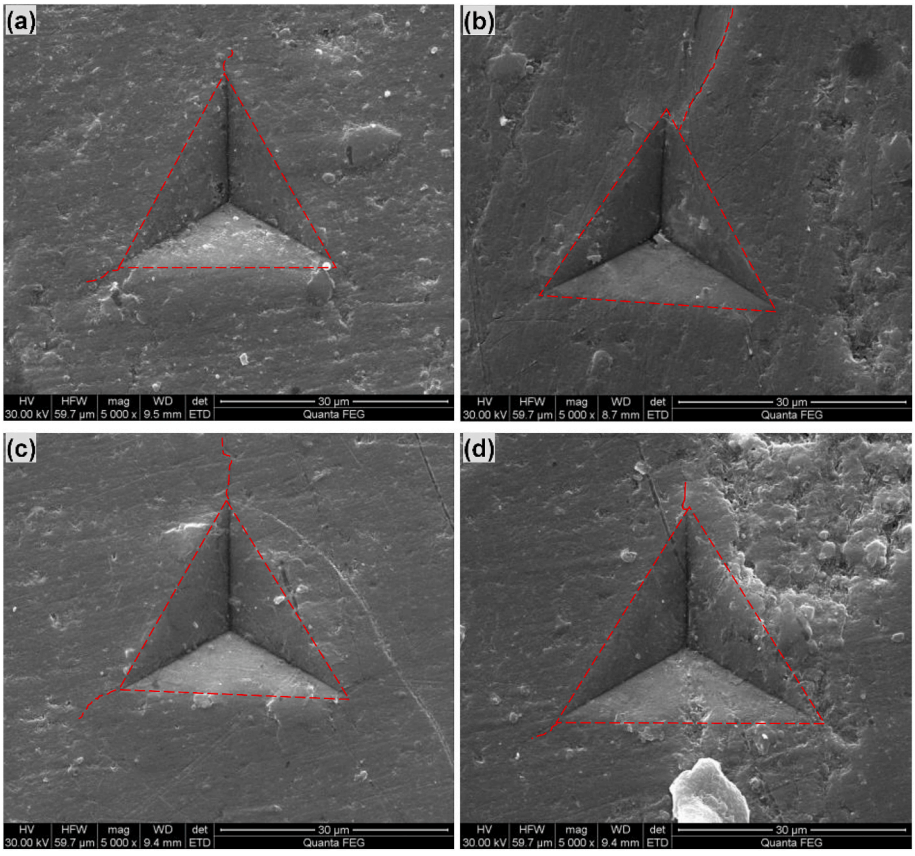


Fig. 8. SEM observation of typical indents located in carbonate-dominated shale matrix.

d when soaking with CO_2. As shown in Fig. 7c and d, the Young's modulus and the hardness decreased to 58.85 and 1.71 GPa (by approximately 11.2% and 18.6% of those of the original shale specimen), respectively. After that, the Young's modulus slightly decreased to 56.50 and 56.66 GPa while the hardness decreased to 1.65 and 1.47 GPa after soaking for 4 and 6 d, respectively. The maximum degradation extents of Young's modulus and the hardness were 14.8% and 30% which were significantly lower than those after soaking with water.

3.3. Typical modes of indents

To further discuss the influence of shale softening behavior on mechanical property, the morphology of residual indents at different locations was characterized by SEM observation after nanoindentation tests. Then, three typical modes of indents were summarized by combining the failure characteristics of a single indent with the load-indentation depth curve and the evolution of micromechanical parameters.

3.3.1. Indents located in shale matrix

Fig. 8 shows the typical SEM images of residual indents located in the carbonate-dominated matrix. It was observed that these indentation areas were characterized by standard triangle shapes, which were accompanied by mico fractures initiating at the tip of the indents and then propagating outward. In this case, the characteristics of the load-indentation depth curves (Fig. 9a) were closely similar to that of the typical experimental curve (Fig. 2d). Meanwhile, Fig. 9b and c shows that the Young's modulus and the hardness underwent an instantaneous initial rising stage and then quickly decreased to a plateau with increasing indentation depth. Additionally, there was an obvious positive correlation between the maximum load and the ultimate micromechanical parameters, as shown in Fig. 9. These differences in the micromechanical strength of the shale surface can be attributed to the differences in the fluid-rock interaction extent, as well as the mineral composition due to heterogeneity.[20] As shown in Figs. 8d and 9, due to the occurrence of shale surface softening which was indicated by the lower Young's modulus and the hardness (see green lines in Fig. 9b and c), the maximum load required at the target indentation depth reduced (see green lines in Fig. 9a) while the area of the residual indents was consequently enlarged (Fig. 8d).

3.3.2. Indents located in vicinity of shale laminas

The shale specimens used in this study were carbonate-rich laminar shale. This laminar structure will further influence the micromechanical property of the shale surface. Fig. 10 shows the SEM images of residual indents located in the vicinity of shale laminas. Due to the accumulation of soft minerals (clay), failure tended to occur on the side of the laminas. As shown in Fig. 10a, the indenter firstly contacted a calcite-dominated mineral and then encountered the laminae with the extension of the contact area. The SEM image of the residual indent shows that an induced fracture was initiated and then attracted by the laminae. Meanwhile, the propagation path of the induced fracture deflected in the laminae, which was attributed to the effect of the directional arrangement of clay minerals. Fig. 10b–d shows that crushed zones with fracture swarm and debris were also formed near the laminas. On one hand, the existence of the fracture swarm improves the permeability of the shale matrix near the fracture surface. On the other, the debris may fall off and accumulate at the pore throats in the proppant package, causing the blockage of fluid flow channels.[26] Additionally, after soaking with water or CO_2, hydration/adsorption-induced fractures were created in the shale laminas. Under this condition, the generation of crushed zone caused the laminas to be compressed. As shown in Fig. 10c and d, one could observe the local closure of hydration/adsorption-induced fractures in the laminas, which would counteract the favorable effect of induced fractures on permeability enhancement to some extent.

Under the effect of shale laminas, the load-indentation depth curve became irregular, as shown in Fig. 11a. As pointed out by arrows in Fig. 11a, one could observe obvious "pop in" phenomena during the loading stage, i.e., the indentation depth significantly increased while the load hardly changed.[37,46] The appearance of "pop in" was closely related to the generation of induced fractures and crushed zones. Additionally, by comparing Figs. 9a and 11a, it was found that the increasing indentation depth during the holding stage caused by shale crept also became larger than those of indents located in the shale matrix. Fig. 11b and c shows that the micromechanical parameters experienced a short initial rising stage (approximately within the initial

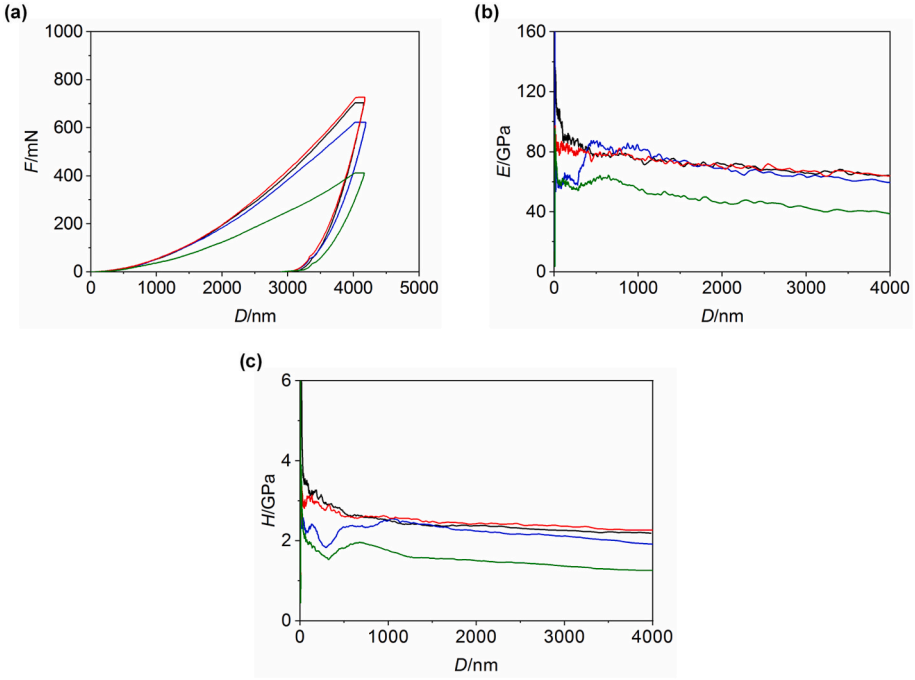


Fig. 9. CSM nanoindentation test results of typical indents located in carbonate minerals: **a** load-indentation depth curve; **b** Young's modulus; **c** Hardness.

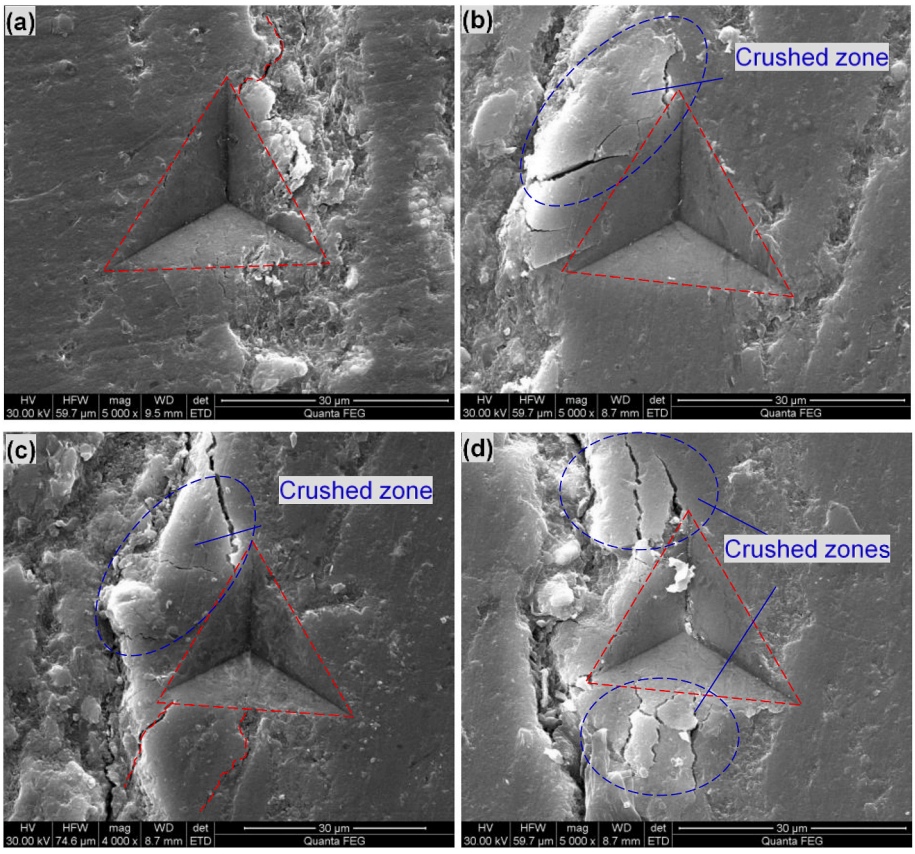


Fig. 10. SEM observation of typical indents located in vicinity of the shale laminas.

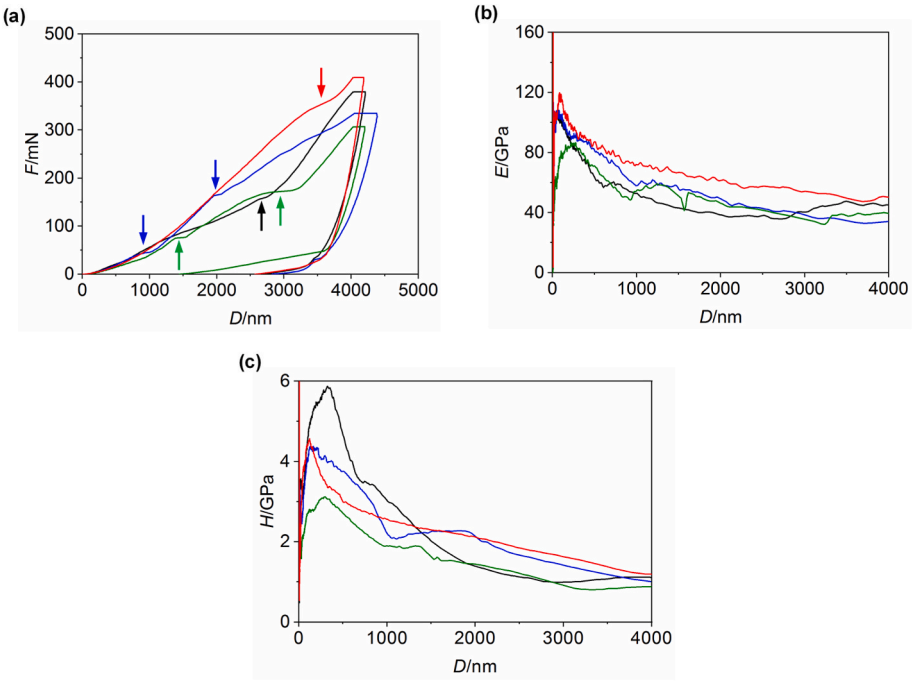


Fig. 11. CSM nanoindentation test results of typical indents located in vicinity of the shale laminas: **a** load-indentation depth curve; **b** Young's modulus; **c** Hardness.

indentation depth of 500 nm) and then gradually decreased. Compared with the curves in Fig. 9b and c, the decreasing rate and amplitude of the micromechanical parameters during the post-peak stage was more significant due to the severe surface failure. Furthermore, when "pop in" phenomena occurred in the load-indentation depth curves, both the Young modulus and the hardness significantly decreased and then rebounded in the meantime.

3.3.3. Indents located in induced fractures

Generally, with the increasing soaking time, more hydration/adsorption-induced fractures tended to be created in the shale specimens, especially when soaking with water. As a consequence, the number of indents located in the induced fractures consequently increased. Figs. 12 and 13 depict the SEM observation results and the CSM nanoindentation test results of indents located in the induced fractures, respectively.

As shown in Fig. 13a, the "delayed loading" phenomena, which was indicated by ultra-low (approximately zero) load during a large range of initial indentation depth, were observed at the beginning of the loading stage in the load-indentation depth curves. According to Hu et al.[40] and Luo et al.,[36] this "delayed loading" phenomenon was attributed to the following two potential mechanisms: (1) the unstable contact between the tip of the indenter and the shale surface, and (2) the incorrect surface detection due to roughness and defects on the testing surface. In this study, the occurrence of the "delayed loading" was closely related to the existence of hydration/adsorption-induced fractures which was the most important indication of shale surface softening. Thus, these data were also considered in statistical analysis.

To some extent, the duration of the "delayed loading" stages was related to the apertures of induced fractures. When the indents were located in the induced fractures with small apertures (Fig. 12a and b), the "delayed loading" phenomena occurred within the range of indentation depth from 0 to approximately 1000 nm (red and black lines in Fig. 13a). With the further increase of indentation depth, the contact area enlarged and the surrounding softer clay matrix was compacted, as shown in Fig. 12a and b. The formation of compaction zones could cause a reduction in the permeability of shale laminas. When the indenter encountered wider induced fractures, the aperture of induced fractures might be locally enlarged due to the pyramid structure of the indenter (Fig. 12c). Although the local extension of induced fracture can improve the permeability of shale laminas, it exacerbates the embedment of proppant. According to the load-indentation depth curves, the maximum loads at the target indentation depth were in the range of 100–260 mN for indents located in induced fractures, obviously lower than those of indents located in shale matrix and vicinity of shale laminas (Figs. 9a, 11a and 13a). Additionally, Fig. 12d shows that the debris entered the shale laminas, which might cause the blockage of the induced fracture.

It is worth noting that when were indents located in induced fractures, both the Young's modulus and the hardness experienced a plateau during the delayed loading stage, as shown in Fig. 13b and c. After that, the micromechanical parameters gradually increased with the further increase of the indentation depth. The rising stage tended to maintain a considerable range of indentation depth. It agreed well with the abnormal evolution tendency of statistical micromechanical parameters after soaking with water for 6 d (green line in Fig. 5a). Thus, this micromechanical response of indents can be used to account for the statistical characteristics of micromechanical parameters in Fig. 5. According to the SEM observation results (Fig. 3a–d), the number and the aperture of hydration-induced fractures increased with the soaking time. Consequently, the probability of indents falling on the fractures tended to increase.

4. Discussion

It has been proved in many previous studies that the occurrence of shale softening is likely to aggravate the proppant embedment, which further causes the loss of fracture conductivity.[16,26,27] Generally, the more severe the shale softening, the more significant reduction of fracture conductivity, consequently the more severe decline in oil/gas production.[25] According to the experimental results of SEM observation and nanoindentation tests, even though the content of swelling clay minerals

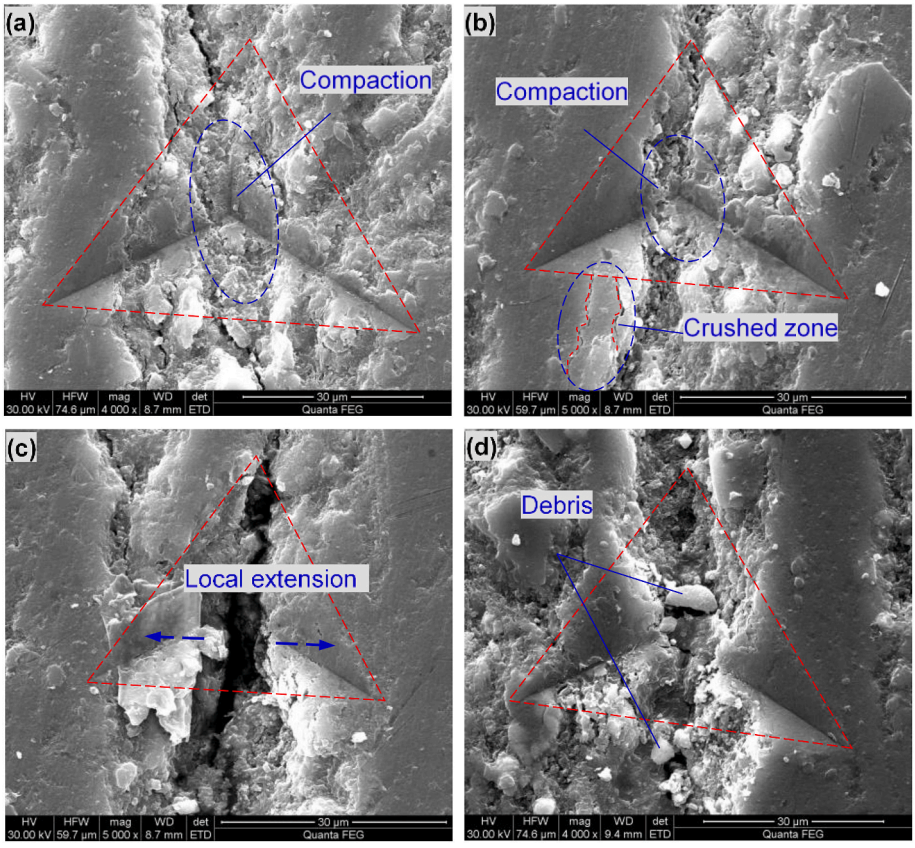


Fig. 12. SEM observation of typical indents located in induced fractures.

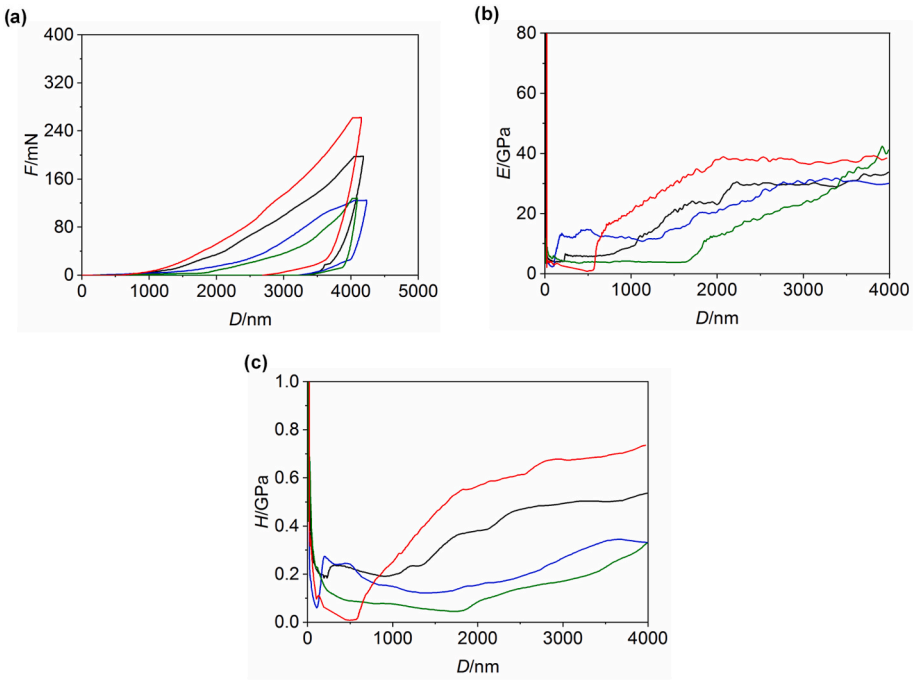


Fig. 13. CSM nanoindentation test results of typical indents located in induced fractures: **a** load-indentation depth curve; **b** Young's modulus; **c** Hardness.

was limited in the studied carbonate-rich shale, the extent of shale softening caused by the hydration effect was severe. The Young's modulus and the hardness of shale decreased by 34.6% and 56.8%, respectively, after soaking with water for 6 d (Fig. 7a and b). The reduction extent of shale surface strength may be more severe in real formations because the duration of hydraulic fracturing and shut-in treatments is generally longer. Thus, it is recommended that the hydration effect in low-clay shale formation should not be neglected. The anti-swelling agent can be considered to lighten the shale softening by alleviating the clay swelling so that the long-term performance of the hydraulic fracture network can be improved. According to the experimental results of Gupta et al.,[47] the increase in the anti-swelling agent (KCl) was beneficial for lowering shale softening. Apart from that, CO_2 fracturing is also conducive to the long-term effectiveness of the hydraulic fracture network from the perspective of alleviating shale softening. As shown in Fig. 7c and d, the degradations of the Young's modulus and the hardness of shale after soaking with CO_2 were lower than that after soaking with water. Furthermore, the degradation extent of micromechanical strength tended to hardly increase after the soaking time exceeded 2 d. This finding complements the advantages of utilizing CO_2 fracturing in shale formation.

Even though some new insight into the shale softening behavior after water/CO_2-rock interaction was provided in this study, there still exist some limitations or issues worthy of in-depth investigation. Firstly, the characterization of shale formations in different areas varies in mineral composition, microstructure, wettability, maturity, etc.[2] The influence of fluid-rock interaction in softening behavior of different shale still needs to be systematically investigated. Secondly, the influence of formation fluid should be considered in future studies. When considering the existence of formation fluid, the interaction between injected fluid (water or CO_2) and rock may be different. For instance, the existence of crude oil changes the wettability of shale, dissolves CO_2 resulting in self-expansion, and influences water infiltration.[48] The existence of initial water can lead to Ion exchange between initial and injected water and further influences the hydration process. Additionally, when the injected CO_2 was dissolved into formation water, the chemical effect of the formed carbonic acid will enhance the mineral dissolution.[20,24] These above-mentioned mechanisms will further influence the shale softening behavior. Thirdly, the shale softening behavior caused by fracturing fluid-rock interaction and its influence on fracture conductivity evolution should be considered in the optimization of shut-in and flowback strategy as well as the prediction of production performance.

5. Conclusion

In this study, the shale softening behavior induced by water/CO_2-rock interaction was investigated by combining the static soaking treatments and the nanoindentation tests. The findings are summarized as follows:

(1) The generation of induced fractures was the dominant softening mode of the studied carbonate-rich shale after the static soaking treatments. The extent of shale softening was closely related to the types of soaking fluids and the soaking time. The alteration of the microstructure of shale specimens after soaking with water was more significant than that after soaking with CO_2. The aperture of hydration-induced fractures increased with the increasing soaking time.

(2) After soaking with water, the micromechanical strength of the carbonate-rich shale significantly decreased. Compared to the origin shale specimen, the Young's modulus decreased by 13.5%, 21.6%, and 34.6% with the soaking time increasing from 2 to 6 d, while the hardness decreased by 27.6%, 29.0%, and 57.6%, respectively.

(3) The shale softening extent of shale after soaking with CO_2 was smaller than that after soaking with water. The Young's modulus and the hardness were ultimately decreased by 14.8% and 30.0% after soaking with CO_2 for 6 d, respectively. Meanwhile, the micromechanical degradation of the shale surface mainly occurred in the first 2 d due to the high penetrability of CO_2.

(4) Due to the existence of induced fractures, indents at different locations show obvious differences in geometry characteristics of residual indents as well as the corresponding response of load and micromechanical parameters with the indentation depth. Particularly, indents located in the vicinity of induced fractures tended to cause crushed zones near the indents and obvious "pop

in" phenomena in the load-indentation depth curves. Additionally, indents located in induced fractures tended to cause the abnormal increase of statistical micromechanical strength with increasing indentation depth.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

This paper was supported by the Chinese Academy of Sciences (KKBE170026), the SINOPEC Group (No.P20049-1; P21039-3; YK-2021-29-2).

References

1 Rana A, Aefaj MK, Saleh TA. Advanced developments in shale inhibitors for oil production with low environmental footprints-A review. *Fuel*. 2019;247:237–249.
2 Jin ZJ, Zhu RK, Liang XP, Shen YQ. Several issues worthy of attention in current lacustrine shale oil exploration and development. *Petrol Explor Dev*. 2021;48(6):1276–1287.
3 Liu BL, Suzuki A, Ito T. Numerical analysis of different fracturing mechanisms between supercritical CO_2 and water-based fracturing fluids. *Int J Rock Mech Min Sci*. 2020;132, 104385.
4 Zou YS, Li N, Ma XF, Zhang SC, Li SH. Experimental study on the growth behavior of supercritical CO_2-induced fractures in a layered tight sandstone formation. *J Nat Gas Sci Eng*. 2018;49:145–156.
5 Wu S, Ge HK, Li TT, et al. Characteristics of fractures stimulated by supercritical carbon dioxide fracturing in shale based on acoustic emission monitoring. *Int J Rock Mech Min Sci*. 2022;152, 105065.
6 Bandara KMAS, Ranjith PG, Haque A, Wanniarachchi WAM, Zheng W, Rathnaweera D. An experimental investigation of the effect of long-term, time-dependent proppant embedment on fracture permeability and fracture aperture reduction. *Int J Rock Mech Min Sci*. 2021;144, 104813.
7 Aliu AO, Xie LZ, He SG. A model to describe hydraulic fracture conductivity decline curve in 3-stages -focus on Longmaxi formation. *J Nat Gas Sci Eng*. 2022;100, 104458.
8 Zhou T, Zhang SC, Yang L, Ma XF, Zou YS, Lin H. Experimental investigation on fracture surface strength softening induced by fracturing fluid imbibition and its impacts on flow conductivity in shale reservoirs. *J Nat Gas Sci Eng*. 2016;36:893–905.
9 Song JL, Xiang DG, Hu DW, Zhou H, Guo DD, Zhang GP. Creep characteristics of a fracturing fluid-softened shale investigated by microindentation. *Int J Rock Mech Min Sci*. 2022;152, 105067.
10 Bai H, Zhang QL, Li ZM, et al. Effect of fracture on production characteristics and oil distribution during CO_2 huff-n-puff under tight and low-permeability conditions. *Fuel*. 2019;246:117–125.
11 Memon S, Feng RH, Ali M, et al. Supercritical CO_2-Shale interaction induced natural fracture closure: implications for $scCO_2$ hydraulic fracturing in shales. *Fuel*. 2022;313, 122682.
12 Xue HQ, Zhou SW, Jiang YL, Zhang FD, Dong Z, Guo W. Effects of hydration on the microstructure and physical properties of shale. *Petrol Explor Dev*. 2018;45(6):1075–1081.
13 Zeng FH, Zhang Q, Guo JC, Zeng B, Zhang Y, He SG. Mechanisms of shale hydration and water block removal. *Petrol Explor Dev*. 2021;48(3):646–653.
14 Lyv Q, Long XP, Ranjith PG, Tan JQ, Kang Y, Wang ZH. Experimental investigation on the mechanical properties of a low-clay shale with different adsorption times in sub-/super-critical CO_2. *Energy*. 2018;147:1288–1298.
15 Liu KQ, Ostadhassan M, Bubach B, Dietrich R, Rasouli V. Nano-dynamic mechanical analysis (nano-DMA) of creep behavior of shales: Bakken case study. *J Mater Sci*. 2018;53:4417–4432.
16 Zhong Y, Kuru E, Zhang H, Kuang JC, She JP. Effect of fracturing fluid/shale rock interaction on the rock physical and mechanical properties, the proppant embedment depth and the fracture conductivity. *Rock Mech Rock Eng*. 2019;52:1011–1022.
17 Lu YY, Ao X, Tang JR, Jia YZ, Zhang XW. Swelling of shale in supercritical carbon dioxide. *J Nat Gas Sci Eng*. 2016;30:268–275.
18 Zou YS, Li SH, Ma XF, Zhang SC, Li N, Chen M. Effects of CO_2–brine–rock interaction on porosity/permeability and mechanical properties during supercritical-CO_2 fracturing in shale reservoirs. *J Nat Gas Sci Eng*. 2018;49:157–168.
19 Ozotta O, Liu KQ, Gentzis T, et al. Pore structure alteration of organic-rich shale with Sc-CO_2 exposure: the Bakken Formation. *Energy Fuels*. 2021;35:5074–5089.
20 Li SH, Zhang SC, Xing HL, Zou YS. CO_2-brine-rock interactions altering the mineralogical, physical, and mechanical properties of carbonate-rich shale oil reservoirs. *Energy*. 2022;256, 124608.
21 Pan Y, Hui D, Luo P, Zhang Y, Zhang L, Sun L. Influences of subcritical and supercritical CO_2 treatment on the pore structure characteristics of marine and terrestrial shales. *J CO2 Util*. 2018;28:152–167.
22 Okamoto I, Li X, Ohsumi T. Effect of supercritical CO_2 as the organic solvent on cap rock sealing performance for underground storage. *Energy*. 2005;30:2344–2351.
23 Zhou JP, Yang K, Zhou L, et al. Microstructure and mechanical properties alterations in shale treated via CO_2/CO_2-water exposure. *J Petrol Sci Eng*. 2021;196, 108088.
24 Zhou JP, Tian SF, Zhou L, et al. Effect of sub-/super-critical CO_2 and brine exposure on the mechanical and acoustic emission characteristics of shale. *J Nat Gas Sci Eng*. 2021;90, 103921.
25 Du JT, Hu LM, Meegoda JN, Zhang GP. Shale softening: observations, phenomenological behavior, and mechanisms. *Appl Clay Sci*. 2018;161:290–300.
26 Li N, Zhang SC, Ma XF, Zou YS, Li SH, Zhang ZP. Thermal effect on the evolution of hydraulic fracture conductivity: an experimental study of enhanced geothermal system. *J Petrol Sci Eng*. 2020;187, 106814.
27 Li N, Zhou T, Zhao MY, et al. Thermal-shock-induced surface softening and conductivity evolution of hydraulic fracture in enhanced geothermal system. *Geothermics*. 2022;102(1), 102405.
28 Liu KQ, Ostadhassan M. Microstructural and geomechanical analysis of Bakken shale at nanoscale. *J Petrol Sci Eng*. 2017;153:133–144.
29 Zhang W, Zhang DX, Zhao JL. Experimental investigation of water sensitivity effects on microscale mechanical behavior of shale. *Int J Rock Mech Min Sci*. 2021;145, 104837.
30 Phani PS, Oliver WC, Pharr GM. Understanding and modeling plasticity error during nanoindentation with continuous stiffness measurement. *Mater Des*. 2020;194, 108923.
31 Liu YW, Liu A, Liu SM, Kang Y. Nano-scale mechanical properties of constituent minerals in shales investigated by combined nanoindentation statistical analyses and SEM-EDS-XRD techniques. *Int J Rock Mech Min Sci*. 2022;159, 105187.
32 Song JL, Xiang DG, Zhao SX, Hu DW, Zhou H, Zhang GP. Shale softening degree and rate induced by fracturing fluid under THMC coupling condition. *J Nat Gas Sci Eng*. 2021;96, 104294.
33 Lu YH, Li YC, Wu YK, Luo SM, Jin Y, Zhang GP. Characterization of shale softening by large volume-based nanoindentation. *Rock Mech Rock Eng*. 2020;53:1393–1409.
34 Liu YK, Yang C, Wang JF, Xiong YQ, Peng PA. New insights into hydration-induced creep behavior of shale: a comparison study of brittle black shale and clayey oil shale at micro-scale. *Mar Petrol Geol*. 2022;138, 105554.
35 Shi X, Jiang S, Wang ZX, Bai B, Xiao DS, Tang MH. Application of nanoindentation technology for characterizing the mechanical properties of shale before and after supercritical CO_2 fluid treatment. *J CO2 Util*. 2020;37:158–172.
36 Luo SM, Lu YH, Wu YK, et al. Cross-scale characterization of the elasticity of shales: statistical nanoindentation and data analytics. *J Mech Phys Solid*. 2020;140, 103945.
37 Shi X, Jiang S, Lu SF, et al. Investigation of mechanical properties of bedded shale by nanoindentation tests: a case study on Lower Silurian Longmaxi Formation of Youyang area in southeast Chongqing, China. *Petrol Explor Dev*. 2019;46(1):163–172.
38 Oliver WC, Pharr GM. An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments. *J Mater Res*. 1992;7(6):1564–1583.
39 Oliver WC, Pharr GM. Measurement of hardness and elastic modulus by instrumented indentation: advances in understanding and refinements to methodology. *J Mater Res*. 2004;19(1):3–20.
40 Hu C, Gao Y, Zhang Y, Li Z. Statistical nanoindentation technique in application to hardened cement pastes: influences of material microstructure and analysis method. *Construct Build Mater*. 2016;113:306–316.
41 Wu YK, Li YC, Luo SM, et al. Multiscale elastic anisotropy of a shale characterized by cross-scale big data nanoindentation. *Int J Rok Mech Min Sci*. 2020;134, 104458.
42 Zhang ZP, Zhang SC, Shi SZ, et al. Evaluation of multi-scale mechanical properties of conglomerate using nanoindentation and homogenization methods:A case study on tight conglomerate reservoirs in southern slope of Mahu sag. *Chin J Rock Mech Eng*. 2022;41(5):926–940.
43 Ketzer JM, Iglesias R, Einloft S, Dullius J, Ligabue R, De Lima V. Water-rock-CO_2 interactions in saline aquifers aimed for carbon dioxide storage: experimental and numerical modeling studies of the Rio Bonito Formation (Permian), southern Brazil. *Appl Geochem*. 2009;24(5):760–767.
44 Cheng P, Zhang CP, Ma ZY, et al. Experimental study of micromechanical properties alterations of shale matrix treated by $ScCO_2$-Water saturation using nanoindentation tests. *Energy*. 2022;242, 122965.
45 Xu JJ, Tang XH, Wang ZZ, Feng YF, Kang B. Investigating the softening of weak interlayers during landslides using nanoindentation experiments and simulations. *Eng Geol*. 2020;277, 105-801.
46 Zhang G, Wei Z, Ferrell RE. Elastic modulus and hardness of muscovite and rectorite determined by nanoindentation. *Appl Clay Sci*. 2009;43:271–281.
47 Gupta I, Rai C, Sondergeld C, Hofmann R. Water weakening: case study from Marcellus and Woodford. In: *SPE/AAPG/SEG Unconventional Resources Technology Conference, Unconventional Resources Technology Conference*. Society of Petroleum Engineers; 2019.
48 Wang FY, Zhao JY. A mathematical model for co-current spontaneous water imbibition into oil-saturated tight sandstone: upscaling from pore-scale to core-scale with fractal approach. *J Petrol Sci Eng*. 2019;178:376–388.